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Manual

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System Visuals

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JetBot System I

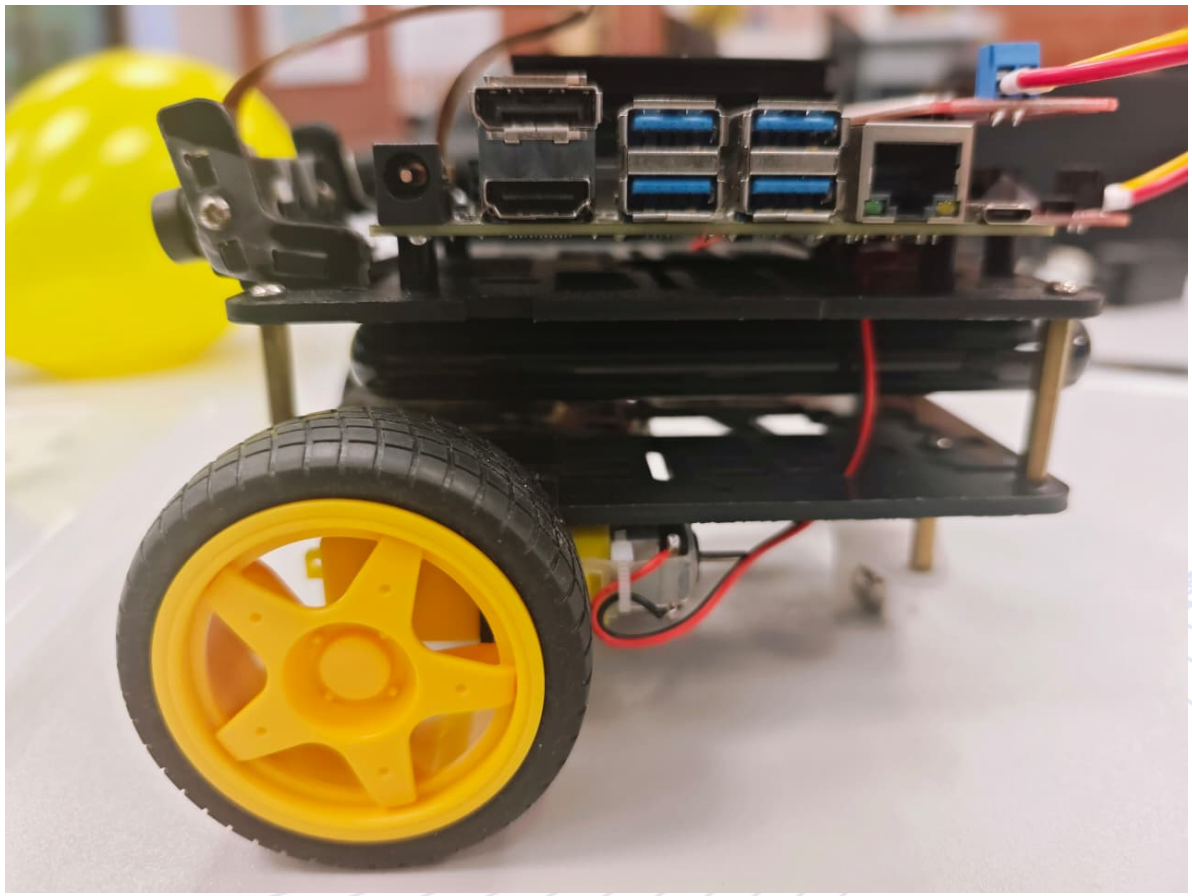


Abbildung: Sytem Side View

JetBot System Top View I

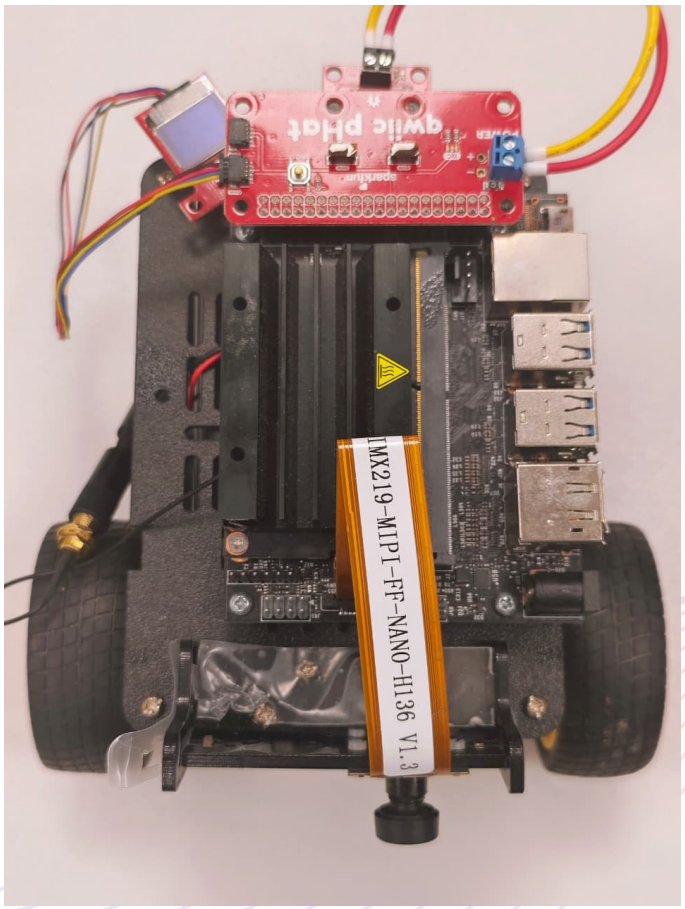


Abbildung: System Top View

JetBot System Front View I

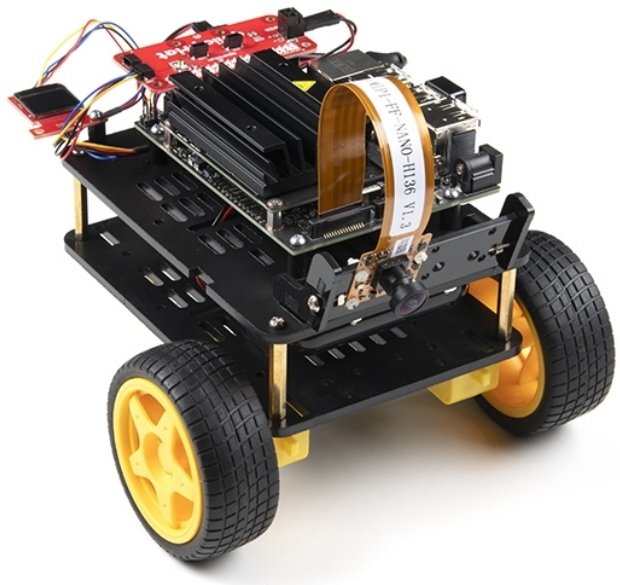


Abbildung: System Front View

JetBot System Back View I

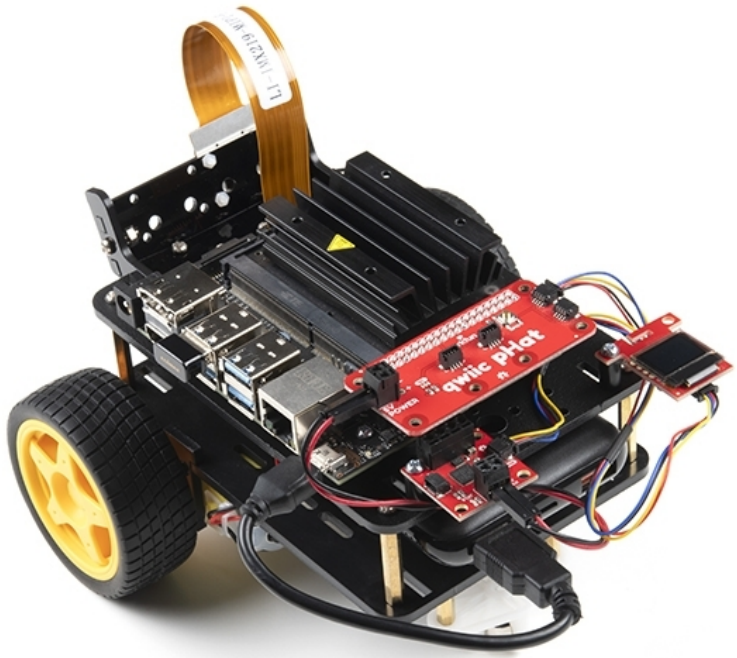


Abbildung: System Back View

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First Steps

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Connections I

1. Insert the pre-flashed 64GB microSD card into the Jetson Nano.
2. Connect the Leopard Imaging IMX219 CSI ribbon to CAM0; ensure contacts face inward.
3. Mount and connect the Qwiic Motor Driver to the Qwiic bus; attach motors.
4. Connect Edimax USB WiFi, keyboard, mouse, and HDMI/DP display.
5. Connect recommended 5V @ 4A regulated power supply and power on.
6. Confirm OS boots and JetBot localization image loads.

First Steps I

1. Power on the JetBot system using a stable 5V–4A power supply.
2. Wait for Ubuntu to fully boot into the desktop environment.
3. Log in with your configured username and password. Username:ros2 Password:ros2
4. Verify WiFi connection (top-right network icon or `nmcli device status`).
5. Open a terminal and confirm camera detection:
   - `dmesg | grep -i camera`
   - `ls /dev/video*`
6. Launch Jupyter Notebook:
   - Navigate to JetBot directory
   - Run `jupyter notebook`

First Steps - Inside System I

1. Open the Ground Localization notebook or application script.
2. Run camera preview cell to confirm live feed.
3. Execute calibration routine if first-time setup.
4. Start localization demo or main control script.
5. Verify:
   - Live camera feed is active
   - Localization status shows “OK”
   - Motors respond to control commands
6. Use UI controls to Start / Stop / Calibrate as needed.

Setup I

- Complete Ubuntu initial setup (locale, user, updates).
- Open a terminal and verify camera is detected (e.g. `dmesg | grep -i camera`).
- Run quick camera test (OpenCV or GStreamer preview).
- Start the JetBot localization demo / Jupyter notebook.
- Perform a short calibration run over representative ground textures.

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Main Function

Main Function I

- Capture continuous ground-facing images via IMX219 camera.
- Extract the visual features from ground texture (keypoints / descriptors).
- Compute frame-to-frame motion using visual odometry (VO).
- Fuse VO into a pose estimator and provide incremental localization updates.
- Output pose and velocity to motion controller for closed-loop navigation.

Help I

- Use the built-in **Help** menu in the UI for quick-start steps and diagnostics.
- Online resources: JetBot GitHub, NVIDIA Jetson forums, SparkFun documentation.
- Local troubleshooting logs: `/var/log/syslog`, `dmesg`, and ROS topic logs.
- Contact support: provide system image version, microSD serial, and captured logs.

Functions I

- **Localize**: continuous pose estimation relative to ground features.
- **Calibrate**: camera intrinsic calibration and ground-texture sampling routine.
- **Record**: save camera frames and computed poses for offline analysis.
- **Play**: replay recorded sessions in the UI for debugging.
- **Export**: export calibration parameters and logs (JSON / NPZ).

Maintenance I

- Monthly: inspect and reseat CSI ribbon and Qwiic connectors.
- Every 3 months: re-run calibration if the robot undergoes mechanical changes.
- Keep software updated: apply tested JetPack / JetBot image updates only.
- Replace power adapter if voltage drops or frequent resets occur.
- Clean camera lens with microfiber cloth; avoid solvents.

Troubleshooting I

- **No camera preview**: reseat FFC ribbon (contacts inward), reboot, check `dmesg`.
- **Low FPS / lag**: verify power supply, reduce resolution, check CPU/GPU utilization.
- **Localization drift**: collect diverse calibration samples, improve lighting, enable feature filtering.
- **Motors jitter**: inspect motor wiring, verify motor driver current limits, update PID gains.
- **Network unreachable**: reconfigure WiFi or use wired Ethernet adapter.

Specifications I

- **Compute**: NVIDIA Jetson Nano Developer Kit (Quad-core ARM A57, 128-core Maxwell GPU).
- **Camera**: Leopard Imaging IMX219, 3280×2464 px, 136° FOV, MIPI CSI-2 (2-lane), up to 21 FPS at full res.
- **Storage**: 64GB microSD pre-flashed image.
- **Power**: 5V recommended, 4A for full AI workloads.
- **Connectivity**: AC8265 WiFi-Card + 6DB Antenna-Networkadapter for Jetson Nano Qwiic I^2C, HDMI cable, SparkFun Qwiic pHAT for Raspberry Pi V2.0, SparkFun Qwiic Motor Driver, SparkFun Micro OLED Breakout (64x48 pixels, Qwiic, I^2C, 10mA, addressable, 3.3V).
- **Mechanical**: Adafruit Height Metal Caster Bearing Wheel (15.7mm wheel diameter, 20mm height) Waveshare Smart Mobile Robot Chassis Kit (includes 4 wheels, 4 TT motors, metal chassis, shock-absorbing features), ISM 433 MHz Antenna (SMA, 2.5 dBi, omnidirectional, flexible, black).

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Next Steps

Next Steps I

- Follow the Report to know more about the system.
- Check for OS version if it is compatible with the system.
- Follow the Readme.md file to know more about the system.
- For more information on JetBot click here.

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Vielen Dank

für Ihre Aufmerksamkeit